

Query Rewriting, Retrieval Framing, and Representational Drift

Why a RAG System May Search for a Different Question Than the User Asked

AI Memory and Reality Drift - Note #4 v1

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The Basic Pattern

Many RAG systems do not search directly with the user's original words. They rewrite ambiguous questions, expand keywords, generate hypothetical answers, decompose complex requests into subquestions, or route the query to a specialized retriever.

These techniques often improve recall. A short question can be converted into language closer to the indexed documents. But the rewritten query is also an interpretation. It selects an intent, supplies implied context, and determines which parts of the question become searchable.

Retrieval can therefore succeed against a transformed question while failing to answer the question the user meant.

When the Query Representation Becomes Operational

A user's request is usually underspecified. The system must infer entities, timeframe, domain, desired evidence, and level of detail. Query transformation makes those assumptions operational before sources are selected.

If the rewrite broadens the question, retrieval may return generic material. If it narrows too aggressively, competing interpretations disappear. If a hypothetical answer is generated first, retrieval can become a search for evidence resembling the model's initial expectation.

How Retrieval Framing Emerges

Stage 1 - User expression. A question arrives with ambiguity, context, and unstated intent.

Stage 2 - Interpretation. A model infers the question's likely meaning.

Stage 3 - Transformation. The query is rewritten, expanded, decomposed, or routed.

Stage 4 - Retrieval. Sources are selected according to the transformed representation.

Stage 5 - Apparent confirmation. Retrieved material supports the framing that determined what could be found.

Semantic Drift and Reality Drift

Semantic Drift occurs when meaning changes across transformations while enough continuity remains for the result to appear faithful. Query rewriting can preserve topic while changing scope, presupposition, temporal frame, or evidentiary burden.

Reality Drift appears when the system repeatedly answers its own stable representations of user intent. The application remains responsive, yet the retrieval environment becomes increasingly coupled to machine-generated questions rather than the problem outside the system.

The Representation Stack

User need becomes natural-language question. The question becomes an inferred intent. Intent becomes one or more retrieval queries. Queries produce a ranked evidence set. The model answers from that evidence and may store the interaction as proof that the framing worked.

The decisive transformation occurs before retrieval. The system does not merely search the index. It first constructs the question that the index will be allowed to answer.

Examples Across AI Systems

Medical search: A symptom question may be rewritten around the most common diagnosis, narrowing the evidence before retrieval.

Enterprise search: A request about a failed process may be reframed as a request for the official procedure.

Research assistance: An exploratory question may be decomposed into assumptions that favor one theoretical vocabulary.

Customer support: Intent routing may send a novel problem into the nearest existing issue category.

Conclusion

Query rewriting is valuable because retrieval requires a usable representation of intent. The risk lies in forgetting that the representation is an inference rather than the question itself.

A grounded system should expose important rewrites, preserve alternative interpretations, and allow the user's original language to return as a constraint. Otherwise the answer can become accurate relative to a question no one actually asked.

Related Phrases and Concepts

- query rewriting
- query expansion
- query decomposition
- multi-query retrieval
- HyDE
- retrieval routing
- intent detection

Semantic Fidelity Project Resources

- [Research Library \(GitHub\)](#)
- [Articles & Essays \(Substack\)](#)
- [Semantic Fidelity Glossary](#)